

# Visualizing Confidence in Delivery Robots: Insights from Two Online Studies

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Figure 1: A simulation of a bottleneck scenario featuring a non-yielding delivery robot. The robot on the right signals its intention to proceed, displaying a green bar indicating "confidence: 98%" suggesting the robot is highly certain about its decision.

## Abstract

The growing deployment of mobile robots for last-mile delivery requires their safe and efficient navigation in urban environments. Although considerable research has focused on enhancing robots' understanding of human movement, optimizing communication strategies remains an underexplored area. Transparent communication of a robot's intentions can help reduce confusion and facilitate safe interactions with nearby pedestrians. This study investigates how visual displays of a robot's confidence in its navigation intentions can enhance human-robot interaction. An initial online study demonstrates that both numerical displays and bar graphs improved interpretability. A second online study finds an effect of confidence communication on participants' perception of the robot's predictability, although this effect did not extend to other measures. These results align with previous research on communicating confidence in human-robot interaction, yet discrepancies arose regarding the methods of communication. Our findings contribute to the development of communication strategies that may enhance interactions between robots and pedestrians.

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CHI EA '25, Yokohama, Japan

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ACM ISBN 979-8-4007-1395-8/25/04

<https://doi.org/10.1145/3706599.3719695>

## CCS Concepts

• **Human-centered computing** → **Empirical studies in visualization**; *Empirical studies in HCI*.

## Keywords

Human-robot interaction, Uncertainty, eHMI, Design, Transparency, Explainability, Trust, Predictability, Safety

## ACM Reference Format:

Ronald Cumbal, Natalia Calvo-Barajas, Marina Escobar-Planas, Alexandros Rouchitsas, and Ginevra Castellano. 2025. Visualizing Confidence in Delivery Robots: Insights from Two Online Studies. In *Extended Abstracts of the CHI Conference on Human Factors in Computing Systems (CHI EA '25)*, April 26–May 01, 2025, Yokohama, Japan. ACM, New York, NY, USA, 7 pages. <https://doi.org/10.1145/3706599.3719695>

## 1 Introduction

Sidewalk Delivery Robots (SDRs) are gaining popularity as companies and local governments explore their potential for last-mile delivery. Exploratory studies on human-robot interactions (HRI) in public environments, however, reveal mixed reactions to their use. A common challenge is the SDRs' lack of effective communication, which often leads to ambiguous navigation perceptions [26]. To enhance safety, efficiency, and comfort, mobile robots must adhere to social norms and employ effective communication strategies [9, 40]. One approach is for robots to convey their confidence or uncertainty during interactions. Research suggests that communicating

confidence has the potential to reduce risks and bridge communication gaps, particularly in collaborative tasks [36]. Nonetheless, further research is needed to determine the best ways to integrate confidence into robotic communication, as outcomes can vary. For example, Selkowitz et al. [37] found that transparently communicating a robot's goals, reasoning, projections, and uncertainty improved trust and understanding in teamwork scenarios, though the impact of uncertainty was less significant.

In social navigation, research has focused on conveying a robot's intentions, with relatively less attention given to transparency in decision-making. In this context, confidence communication represents how certain the robot is about its ability to navigate shared spaces. This aspect becomes particularly important when both the robot and a pedestrian need to traverse the same area. By conveying its confidence level during these interactions, the robot can help pedestrians understand its behavior, leading to smoother interactions, i.e., smoother negotiation of shared spaces, and reducing the risk of hesitation or near misses [22]. In particular, we examine how communicating confidence influences pedestrian behavior, particularly in terms of decision accuracy, response time, and perceived safety. Our findings indicate that the effects of communicating confidence are consistent with previous research in HRI. However, we note discrepancies in the effectiveness of the methods used to convey this information.

## 2 Related Work

Delivery robots pose significant challenges related to pedestrian interactions in public spaces, especially concerning pedestrian safety. Gehrke et al. [14] found that robots frequently disrupt the travel patterns of pedestrians and cyclists. Similarly, Van Mierlo et al. [44] reported that while many individuals ignored the presence of sidewalk delivery robots, certain groups, particularly dog owners, actively avoided them. Weinberg et al. [45] reported that delivery robots in a Pittsburgh pilot caused distractions and obstructions, however, they also reported positive interactions. These findings align with a study in Estonia in which pedestrians found the robots charming [13], suggesting a mixed public perception.

To integrate seamlessly into human environments, delivery robots must navigate social contexts effectively and communicate in ways that enhance user experience [7, 20, 38]. Research has explored implicit [10, 41] and explicit [6, 17] communication strategies to facilitate navigation. Weinberg et al. [45] demonstrated how changes in robot communication, from eye shapes and polite phrases to non-linguistic sounds and written directives, increased pedestrian engagement. External Human-Machine Interfaces (eHMIs) have emerged as communication mechanisms for autonomous vehicles (AV) to facilitate such interactions, however, their use in robots remains limited despite their significant potential [26]. While eHMIs are mostly designed to communicate a robot's intentions [39], safe interactions also require an explanation of why a robot acts in a certain way [4, 19, 49]. This interest in explainable robots has led to a growing body of work on interfaces that communicate decision-making processes [11, 16, 35], highlighting the diverse public reactions to delivery robots and the need to improve communication strategies for safer interactions.

The clear communication of a robot's intentions fosters transparency and human agency, both of which are highlighted by Ethical AI regulations as crucial factors for building trust [43]. Abrams

et al. found that acceptance of delivery robots decreases when they are perceived as threatening [1], emphasizing the role of trust in public interactions. Therefore, robots should prioritize features that improve user understanding and effectively communicate their intentions to foster trust and acceptance [9]. Providing explanations can further improve comprehension of a system's actions and foster transparent communication [21, 37]. Schömbbs et al. [36] demonstrated that the type of information shared significantly impacts user judgments and decision-making, raising questions about how best to display transparency.

An effective approach to enhancing transparency is the communication of confidence levels in decision-making, as it promotes active engagement and supports informed decisions [36]. For instance, Yee et al. showed that confidence displays in an AI predicting allergens influenced user decisions and recommended concise formats to ensure quick reactions [48]. However, despite these insights, research on transparent communication in delivery robots remains limited. Given that transparency improves understanding of a system's actions, conveying confidence could further enhance interactions and facilitate safer navigation in shared spaces.

## 3 Research Objectives

This research explores how transparent communication in delivery robots can improve negotiation in shared spaces. Specifically, it examines **how visualizing a delivery robot's confidence levels affects pedestrian interactions**. Here, confidence refers to the robot's certainty about its next action. For instance, a message like "*I will go ahead*" with *low-confidence* suggests uncertainty about the scanned path ahead, whereas the same intention with *high-confidence* signals the robot predicts a smooth trajectory without stopping for the pedestrian. To address the research question, we conducted two online surveys. The first survey focuses on (1) evaluating the *design of eHMIs* used to convey a delivery robot's confidence and (2) evaluating how *pedestrians interpret* the visualization of different confidence levels. The second survey includes an interactive task designed to assess how communicating a delivery robot's confidence affects pedestrians' decision-making, specifically on (3) assessing how confidence communication affects navigation decision *latency* and *accuracy* and (4) measuring pedestrians' perceptions of the robot's *safety*, *reliability*, and *predictability*.

## 4 Simulation Environment

The simulated sidewalk delivery robot and environment were modeled using Gazebo software, with the robot based on a pre-designed Husky model<sup>1</sup>. The robot measured 0.85 m in height, 1 m in length, and 0.57 m in width, consistent with typical SDR dimensions [29], and featured a display on its front. The simulation recreated a common scenario where a pedestrian and robot encounter each other on a shared sidewalk of the standard  $\approx 1.5$  meter width<sup>2</sup>, resulting in a *bottleneck* where only one could proceed (see Figures 1 and 3 for the simulated environment). The remaining scenario was populated with pre-built models from the Gazebo repository. The online surveys were developed using the jsPsych library and hosted on the Cognition platform. Participants were recruited from Prolific.

<sup>1</sup><http://models.gazebo-sim.org/husky/>

<sup>2</sup>ADA Compliance Guidelines for Sidewalks

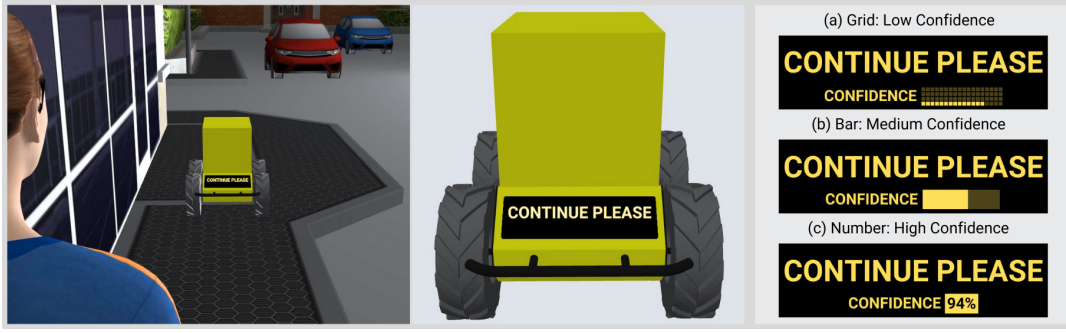


Figure 2: Initial visualizations concepts of a delivery robot's confidence levels: (a) Grid: shows a 25x4 cell array, (b) Bar: shows a progress bar and (c) Number: shows confidence as an integer percentage. The study stimuli included these three concepts at low, medium, and high confidence levels. Participants viewed each stimulus in a similar format to the left two panels.

## 5 Online Survey 1: Evaluating the Visualization of Confidence on a Robot

Research on eHMIs has predominantly focused on communicating intention using visual cues. The preferred type of eHMI often depends on the interactive scenario, but studies suggest that displays are generally favored [26]. Within displays, text is considered a clearer method for conveying intentions. Icons or symbols, while advantageous for being language-independent, may require prior experience for accurate interpretation [5]. This study, therefore, uses *text* and an *arrow symbol* to communicate the robot's intention.

While research on communicating confidence in robots remains limited compared to other fields, established methodologies for visualizing uncertainty in scientific communication exist [8, 32]. Based on previous studies [3, 25, 36], we evaluate three visualization alternatives—a number, a bar graph, and an icon grid—to communicate the delivery robot's confidence in realizing its intention, as shown in Figure 2. We avoided using different colors to prevent inducing user preferences, bias, cultural effects, and to enhance accessibility [24, 42]. In this survey, only the text version was used to display intention (i.e., “Continue Please” [5]) to keep the variable constant, focusing solely on the display of confidence. Drawing on prior research [30, 46], we categorized confidence into three levels: Low (0 – 30%), Medium (30 – 80%) and High (80 – 99%). We created a total of nine images showing three distinct values within each confidence category. The three values for each confidence level were selected by choosing the median of each level range and two numbers equally spaced around this median (i.e., Low: 11 – 15 – 19%; Medium: 51 – 55 – 59%; High: 86 – 90 – 94%).

### 5.1 Procedure

Participants initially provided their age, gender, attitude toward robots (5-point Likert scale) [18], trust in technology (7-point Likert scale) [31, 36], and experience with delivery robots (5-point scale from “None” to “Expert”). They were also shown a reference image and definition of sidewalk delivery robots [29]. Using a within-subject approach, participants were then asked to estimate the confidence level displayed on the robot's screen using the nine previously defined images. At the end, participants were asked to suggest improvements to the designs through a comment box.

### 5.2 Results

Participants were screened for English language proficiency and residence in Sweden. A total of 30 participants completed the survey, though data from 4 participants were excluded for failing attention checks. Of the final 26 participants, 12 identified as female and 14 as male, with an average age of 34.88 years. Results show that the grid display produced the highest absolute error ( $er$ ) in participants' estimation of the robot's actual confidence value (Low  $M_{er} = 18.57$ , Medium  $M_{er} = 9.192$ , High  $M_{er} = 12.307$ ), while the bar display (Low  $M_{er} = 7.615$ , Medium  $M_{er} = 3.153$ , High  $M_{er} = 5.884$ ) was closest in matching the accuracy of the numerical display ( $M_{er} = 0$  at all displayed confidence levels). Qualitative analysis of participant suggestions indicated that seven participants preferred numerical displays for confidence, while five raised concerns about the clarity of the information presented by SDRs. Furthermore, four participants recommended a more “friendly” text, three expressed a preference for not displaying confidence levels at all, and three suggested modifications to the size and color of the display.

## 6 Online Survey 2: Evaluating Pedestrians' Decision-making with an Interactive Task

The second survey assessed participants' decision-making latency and accuracy. Participants viewed video stimuli of the bottleneck scenario and pressed a key once they were confident about whether the robot would yield the right of way. A  $2 \times 2 \times 2 + 2$  factorial design was used to examine three factors: *message coding* (arrow vs. text), *robot intention* (yielding vs. not yielding), and *confidence communication* (absent vs. present), along with a baseline condition where no information was displayed (*blank display*). This design follows similar methodologies for the evaluation of eHMIs in autonomous vehicle research [28].

Based on the first survey results, the display was updated with a bar and numerical representation, enlarged for better legibility, and color-coded to indicate confidence levels (e.g., green for high confidence in Figure 3). The text coding included the phrase “I will go ahead” to indicate the robot's claim of the right of way, while a green downward arrows was used for yielding and a yellow upward arrow for not yielding, following Rettenmaier et al. [34]. For the displayed confidence value, we selected a high level of confidence,



**Figure 3: Left area depicts the displays that the delivery robot employed in the simulation, in accordance with the experimental factorial design used in this study. Right section shows the geometric configuration of the bottleneck scenario used for the human-robot interaction.**

98%, as operational robots typically make decisions with higher levels of confidence. We avoided using a perfect confidence level, i.e. 100%, to avoid perceptions of overconfidence [36].

In the simulation (Figures 1 and 3), a delivery robot approached a bottleneck while a pedestrian stood 1.5 meters away from the bottleneck area (Point E in Figure 3). As the robot approaches a distance of 3 meters from the bottleneck (Point B), i.e. the interaction phase, it reduces its speed to 5km/h and activates its display to present the relevant information—in the conditions where information is shown. In the yielding conditions, the robot stopped 1 meter from the bottleneck (Point C); in the non-yielding condition, it continued to the end of the bottleneck (Point D). The display size was validated to ensure a visual angle of at least 20 minutes of arc (MOA) at 25 meters [34], and all videos were recorded at a resolution of 1620 x 960 pixels.

## 6.1 Procedure

Participants completed the same pre-questionnaire used in the first study, providing demographic data, attitudes and experience with robots, and their disposition to trust technology. They were shown an image and definition of sidewalk delivery robots. Then, they were instructed that their task was to decide whether to walk or stay when encountering the delivery robot in the videos. To perform the task, participants were instructed to press the M key to *move* and the S key to *stay* in place. The system automatically logged the first key pressed and its timestamp. Before the main experiment, participants completed two practice rounds featuring a robot without a display (i.e. no front screen). For the main experiment, participants were always first shown the baseline *blank display*, as recommended by prior research [28], followed by randomized trials based on the factorial design. After each video, participants self-reported their perceived safety, robustness, and predictability of the robot. At the end of the study, participants were invited to provide feedback, particularly suggestions for improving the design of the SDR's display, through an open comment box.

## 6.2 Measurements

We collected both objective and subjective measures to evaluate the effectiveness of the robot's communication strategies. Regarding **objective measures**, we assessed *Effectiveness* as a binary variable –‘correct’ or ‘incorrect’–, indicating whether the strategy influenced the user's behavior as intended. Specifically, the communication

was considered effective if the user stayed in place when the robot signaled it would not yield and began moving when the robot indicated it would yield. Additionally, *Reaction Time* was measured as the duration it took for the user to make a decision in response to the robot's display. For this measure, we considered only participants for whom the communication was effective. Concerning **subjective measures**, participants assessed their experience with the robot using a 5-point scale<sup>3</sup>, evaluating the following dimensions: *safety*, *robustness* and *predictability* (adapted from [2, 47]).

## 6.3 Results

A total of 56 participants completed the interactive online survey, with 24 identifying as female and 32 as male. The average participant age was 30.23 years ( $SD = 7.5$ ). Screening criteria included English proficiency, Swedish location or residence, and ensuring participants used a large screen to accommodate the videos, i.e. at least 1620 x 960 pixels. The mean duration of the study was 9.31 minutes ( $SD = 2.9$ ). When assessing participant's attitudes toward technology, they reported a general moderate attitude toward sidewalk delivery robots ( $M = 2.55; SD = 0.7$ ), little to no previous experience with SDRs ( $M = 0.25; SD = 0.5$ ) and moderate to low trust in technology ( $M = 2.35; SD = 0.8$ ).

**6.3.1 Objective Measurements.** We initiated the analysis of the objective measures by examining **effectiveness** with a Cochran's Q test. The test showed a significant effect of the display on participants' behavior ( $X(7) = 29.17, p < .001$ ). To further assess which factors influence the likelihood of participants behaving as intended, we implemented a binomial logistic regression model with *robot intention*, *message coding*, and *confidence communication* as predictors. Our results indicated that robot intention had a main effect on *effectiveness* (Odds Ratio = 2.81, 95% CI [1.47, 5.38],  $z = 3.122, p = .002$ ) indicating that participants were more likely to react accurately when the robot yielded the right of way compared when it did not. We also found a trend of message coding toward significance (Odds Ratio = 1.50, 95% CI [0.997, 2.25],  $z = 1.948, p = .051$ ) indicating that participants were more likely to interpret the robot's intention when the message was coded as text than as arrow. We did not find any significant effect of *confidence communication* on the *effectiveness* (Odds Ratio = 1.44, 95% CI [0.98, 2.12],

<sup>3</sup>Questionnaire: “Please rate your impression of the delivery robot on these scales”: Threatening - Safe; Unreliable - Reliable; Unpredictable - Predictable.

$z = 1.858$ ,  $p = .063$ ). The model did not reveal any significant interaction effect, indicating that the interaction between robot intention, message coding, and confidence communication did not predict *effectiveness*. See Table 1 for the frequency of cases in which participants effectively behaved according to the robot’s intention.

To assess the effects of our dependent variables on **reaction time**, we filtered our data to include only data points in which the communication was effective. Since the videos for the yielding and non-yielding robot intentions had slightly different durations, resulting in different time windows to make a decision across conditions, we normalized the data to account for these differences. A non-parametric mixed linear model revealed a main effect of robot intention on reaction time ( $\beta = 0.091$ ,  $SE = 0.024$ ,  $z = 3.787$ ,  $p < .001$ ) indicating that users took longer to make a decision when the robot yielded the right of way compared to when it did not. The model did not reveal any main effect on message coding ( $\beta = -0.038$ ,  $SE = 0.024$ ,  $z = -1.563$ ,  $p = .118$ ) or confidence ( $\beta = 0.007$ ,  $SE = 0.025$ ,  $z = 0.266$ ,  $p = .790$ ). We did not find interaction effects between the predictors.

**6.3.2 Subjective Measurements.** A Shapiro-Wilk test indicated that none of the dependent variables for subjective measures followed a normal distribution. Consequently, non-parametric tests were employed for further data analysis. Post-hoc analyses were carried out with Bonferroni correction.

To assess participants’ perceived *safety*, *robustness*, and *predictability*, we constructed three mixed linear regression models. Considering *safety*, the results showed that the robot yielding intention had a significant effect on the reported participant’s **safety** ( $\beta = 1.097$ ,  $SE = 0.165$ ,  $z = 6.636$ ,  $p < .001$ ). Message coding ( $\beta = 0.180$ ,  $SE = 0.161$ ,  $z = 1.117$ ,  $p = .264$ ) and confidence communication ( $\beta = 0.288$ ,  $SE = 0.161$ ,  $z = 1.795$ ,  $p = .073$ ) did not show significant main effect.

Regarding **robustness**, results showed that the text display ( $\beta = 0.380$ ,  $SE = 0.156$ ,  $z = 2.437$ ,  $p = .015$ ) and the presence of the confidence level ( $\beta = 0.382$ ,  $SE = 0.155$ ,  $z = 2.459$ ,  $p = .014$ ) had both a significant effect on the participant’s *robustness* report. Indicating that text and displayed confidence improved the perceived robustness of the system. Nevertheless, the robot intention ( $\beta = 0.267$ ,  $SE = 0.160$ ,  $z = 1.671$ ,  $p = .095$ ) did not show significant main effects. No significant interaction effects were observed between the predictors.

Results indicated a main effect of message coding on **predictability** ( $\beta = 0.540$ ,  $SE = 0.179$ ,  $z = 3.023$ ,  $p = .003$ ), suggesting that text had a positive effect compared to arrows. We also found a significant positive effect of confidence communication on *predictability* when it was present ( $\beta = 0.380$ ,  $SE = 0.178$ ,  $z = 2.138$ ,  $p = .032$ ). However, the model did not show a significant effect of robot intention on *predictability* ( $\beta = 0.279$ ,  $SE = 0.183$ ,  $z = 1.524$ ,  $p = .127$ ).

## 6.4 Participant Feedback

We performed a qualitative analysis of participants’ open-ended responses, which primarily focused on improving the clarity of the robot’s intentions. Five participants suggested using a larger display for better visibility. Eighteen comments addressed the use of *arrows*, with many finding them confusing, especially when used to indicate

**Table 1: Contingency table showing the relationship between message coding, robot intention, confidence, and effectiveness. The values represent the frequency of each combination of variables in the study. Neg.=Negative, Pos.=Positive**

| Message | Robot intention | Confidence | Effectiveness |      |
|---------|-----------------|------------|---------------|------|
|         |                 |            | Neg.          | Pos. |
| arrow   | not yielding    | present    | 30            | 20   |
|         |                 | absent     | 25            | 25   |
|         | yield           | present    | 16            | 30   |
|         |                 | absent     | 13            | 35   |
| text    | not yielding    | present    | 25            | 25   |
|         |                 | absent     | 25            | 25   |
|         | yielding        | present    | 15            | 32   |
|         |                 | absent     | 16            | 33   |

that the robot was stopping. Some participants proposed replacing arrows with universal symbols for stationary intentions. *Text* displays were more popular, with thirteen comments suggesting improvements. Seven participants emphasized making the robot’s messages more polite, while others preferred clearer, user-centric instructions, such as participant 43’s suggestion: “*Make them more explicit, such as ‘please move’.*” Additionally, some participants recommended combining text with arrows to enhance comprehension. *Confidence displays* received fewer comments, with mixed opinions. Some participants appreciated the confidence level indicator and recognized its potential utility. Participant 107 noted: “*Having a confidence meter may help walking civilians to analyze the situation better.*” Conversely, a similar number of participants found the confidence display confusing or unnecessary. For instance, participant 140 mentioned: “*the level of confidence doesn’t make much of a difference for the person who encounters the robot.*” Additional suggestions included using sounds and lights to draw attention and accommodate individuals with impairments. Four participants also recommended using different colors to enhance comprehension of instructions.

## 7 Discussion

Results from the initial survey indicated that the grid display resulted in the highest rate of interpretability errors. These findings contradict previous research on the efficacy of icon arrays in communicating probabilities [32, 36], likely due to variations in the context of the interactions. The second survey showed that confidence communication had no significant impact on most measured variables, except for a positive effect on predictability. This pattern was reflected in participants’ feedback, with only a few commenting on the use of confidence in the display and offering mixed opinions. Despite efforts to improve transparency in autonomous robots—especially in shared spaces where pedestrians and robots must negotiate the right of way—the connection between the displayed text and the confidence level may not have been entirely clear. In social settings, pedestrians naturally navigate shared spaces by predicting and responding to each other’s movements. However, when mutual understanding is lacking, interactions can result in hesitation or near misses [22]. We expected greater transparency to reduce these issues, but our findings did not support this. The

disparity in our results, along with prior studies suggesting that visualizing uncertainty can cause confusion or cognitive overload [15, 27], highlights the importance of carefully selecting the information conveyed in human-robot interactions. Transparency must be context-dependent, ensuring that communication methods are both effective and intuitive. Even the well-established use of arrows to conveying intent produced mixed results. This sentiment was reflected in 18 participants indicating that the arrows were confusing or counterintuitive. Participants were more accurate when observing the robot yield rather than continue moving, suggesting that combining motion with visual communication may enhance clarity. This aligns with research on autonomous vehicles [23] and supports the need for multimodal designs to accommodate diverse user needs, including those with impairments [12].

Interpretability is another crucial factor to consider in this context. Many participants struggled to understand the robot's visual displays, especially arrow-based coding. This difficulty was likely exacerbated by participants' lack of familiarity with SDRs. In contrast to autonomous vehicles, most individuals lack familiarity with mobile robots, which hinders their ability to interpret the visual information presented by these systems effectively. For instance, in research on autonomous vehicles, many studies have leveraged common communication mechanisms between cars and pedestrians, replicating subtle cues like gaze and head movements from drivers to signal yielding behavior at crosswalks. Such forms of communication are not instinctive for individuals interacting with delivery robots, thereby complicating the communication of the robots' decision-making processes. Extended exposure to delivery robots may improve interpretability, though research on lighting-based vehicle deceleration suggests that familiarity might also be counterproductive, as once participants became familiar with novel mechanisms, they exhibited slower response times in detecting the same motion changes [33].

**7.0.1 Limitations.** Using simulations to study human behavior with robots has inherent limitations, as they may not elicit the same responses as real-world interactions. For example, auditory cues, such as robot sounds, which can signal status and intentions, were absent [47]. Additionally, this study tested a limited range of real-world scenarios. Encounters with delivery robots occur in diverse contexts, each with unique effects that were beyond the study's scope. Moreover, we focused only on robots with high confidence levels to ensure consistent comparisons, but future research should explore reactions to robots operating at varying confidence levels. Finally, this study's participants were limited to Swedish residents, who generally held neutral attitudes toward SDRs and technology, with minimal prior experience. While this may have muted the impact of confidence communication, the findings remain relevant given delivery robots' limited deployment to specific regions. As their usage expands, further research can refine these insights for broader applications.

## 8 Conclusion

This study aimed to evaluate how the communication of confidence through different visualization mechanisms affects participants' perceptions and behaviors when encountering a delivery robot. Our findings highlight the complexities involved in presenting

confidence and intent in sidewalk delivery robots. While the results indicated a positive effect of confidence communication on predictability, this effect did not extend to other objective and subjective measures. Participants shared mixed opinions about the usefulness of confidence displays, underscoring the need for careful consideration of the information conveyed in human-robot interactions. Furthermore, the increased interpretability errors associated with the grid display and the confusion arising from arrow-based message coding highlight the necessity of carefully designing interfaces for different contextual interactions. Integrating the robot's motion with its displayed intentions may enhance clarity and effectiveness, as supported by previous research on autonomous vehicles. However, the lack of familiarity with SDRs among participants suggests that additional training or exposure to these communication mechanisms is necessary.

## Acknowledgments

This research was supported by the Horizon Europe EIC project SymAware under the Grant Agreement No. 101070802. We would like to express our gratitude to Didem Gürdür Broo for her support and contributions to this research.

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